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Amorphous calcium phosphate nanowires prepared using beta-glycerophosphate disodium salt as an organic phosphate source by a microwave-assisted hydrothermal method and adsorption of heavy metals in water treatment

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Amorphous calcium phosphate nanowires (ACPNWs) were prepared using calcium chloride as the calcium source and β-glycerophosphate disodium salt (BGP) as the phosphate source by a microwave-assisted hydrothermal method. The effects of the hydrothermal temperature and concentrations of BGP and $CaCl_2$ on the morphology and crystalline phase of the product were investigated. The as-prepared products were characterized by X-ray powder diffraction (XRD), transmission electron microscopy (TEM), and scanning electron microscopy (SEM). The possible formation mechanism of the ACPNWs is proposed. The experimental results indicate that the as-prepared ACPNWs exhibit large adsorption capacities for heavy metal ions (Cd^{2+}, Cu^{2+}, Pb^{2+}, and Zn^{2+}) and a highly selective adsorption activity for Pb^{2+} ions in an aqueous solution in the pH range from 4.5 to 8.5, implying that ACPNWs are a promising adsorbent for applications in water treatment.

Introduction

Nowadays, considerable attention has been paid to the removal of heavy metal ions from wastewater because of their growing threat to plants, animals and human beings due to their bioaccumulation and high toxicity. Thus, an easy, efficient and rapid removal technology to remove toxic heavy metals from waste water is urgently needed.[1,2] Various conventional technologies, including chemical precipitation, ion exchange, membrane filtration, coagulation–flocculation, flotation electrochemical treatment and magnetic assistance and adsorbents, have already been demonstrated as versatile approaches to remove heavy metal ions from aqueous solution.[3–6] Among these methods, adsorption of heavy metal ions using various adsorbents for water treatment appears to be very promising, which can be attributed to the advantages of low cost, high efficiency and ease of operation.

Calcium phosphate materials, as common adsorbents for heavy metals, have attracted a lot of attention in the field of water treatment. Various calcium phosphate adsorbents with different physicochemical properties, including hydroxyapatite nanosheet-assembled microspheres,[7] three dimensionally ordered macroporous hydroxyapatite,[8] hydroxyapatite particles,[9,10] hydroxyapatite hierarchical hollow microspheres,[11] apatite II™ (a biogenic hydroxyapatite),[12,13] and magnetic hydroxyapatite nanoparticles,[14] have been synthesized and investigated for potential applications in the removal of heavy metals. Stötzel *et al.*[15] found that the capacity for metal ion adsorption by hydroxyapatite increased with decreasing crystallinity and increasing specific surface area. Therefore, ACP nanostructured materials are promising adsorbents for the removal of heavy metal ions.

ACP is amorphous in structure and contains 10–20 wt% of bound water. A roughly spherical cluster of ions with a diameter of 9.5 Å was proposed for the basic structural unit of ACP with the composition of $Ca_9(PO_4)_6$.[16–19] The spherical cluster is proposed as the first nuclei for the crystallization of hydroxyapatite.[20] However, ACP as a transient phase will transform to crystalline hydroxyapatite in aqueous solution without stabilization.[21] The stability of ACP in aqueous solution is usually determined by a variety of factors such as the additive, temperature, pH value, and ionic strength. ACP can be stabilized in an aqueous solution using appropriate additive molecules and ions to prevent the transformation from ACP to hydroxyapatite.[22,23] Compared with well-crystallized hydroxyapatite, ACP is bioactive with a better biodegradability and can promote osteoblast adhesion and osteoconductivity.[24,25] In addition, ACP nanostructured materials usually have nanosized pores and channels and a high specific surface area for efficient adsorption of heavy metal ions.

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Several strategies, including the protein/polymer-assisted method,[26,27] dry-mechanochemical method,[28] microwave-assisted hydrothermal method,[23,29] and others,[30] have been developed to prepare ACP nanostructured materials. Among these methods, the microwave-assisted hydrothermal strategy for the rapid synthesis of nanostructured materials has attracted considerable interest due to its advantages of rapidness, high efficiency, low cost and energy saving nature.[31] ACP nanostructured materials including porous nanospheres,[23] porous microspheres/hollow microspheres,[29,32,33] and nanoparticles[34] have been obtained by the microwave-assisted method.

Phosphorus-containing biomolecules as the biocompatible organic phosphorus source have obvious advantages over an inorganic phosphate source. In previous studies, calcium phosphate materials with various morphologies, including ACP nanospheres,[23] ACP mesoporous microspheres,[32] hydroxyapatite nanorods and nanowires,[35] and hierarchical hollow hydroxyapatite microspheres,[36,37] have been prepared using different phosphorus-containing biomolecules by the microwave-assisted method. Compared with an inorganic phosphorus source, the use of phosphorus-containing biomolecules as the organic phosphorus source has obvious advantages, for example, they are biocompatible and essentially non-toxic; the phosphate ions form in a progressive manner by hydrolysis, avoiding the fast nucleation and disordered growth of calcium phosphates; the hydrolysis conditions of biomolecules can be used to control the morphology, size, and structure of the product; and the biomolecules can regulate the crystal growth of the product.[36]

In this study, we report a microwave-assisted hydrothermal synthesis of amorphous calcium phosphate nanowires (ACPNWs) using calcium chloride as a calcium source and β-glycerophosphate disodium salt (BGP) as a phosphorus source. BGP can be hydrolyzed by heat treatment to release phosphate ions, thus acting as a biocompatible organic phosphate source. BGP molecules can also regulate the growth of calcium phosphate. In this work, the effects of the hydrothermal temperature and concentration of BGP and $CaCl_2$ on the morphology and crystalline phase of the product were investigated. The formation mechanism of ACPWs is discussed. The removal capability of ACPWs for metal ions (Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+}) in aqueous solution is studied. The as-prepared ACPNWs have a high surface area and a good adsorption capability for heavy metal ions and a high selective adsorption activity for Pb^{2+} ions, and thus they are a promising adsorbent for water treatment.

Experimental section

Materials

β-Glycerophosphate disodium salt pentahydrate (glycerol-2-phosphate disodium salt, BGP, $C_3H_7Na_2O_6P\cdot 5H_2O$, purity > 97%) was obtained from Sangon BioTech (Shanghai) Co. Ltd. Anhydrous calcium chloride (analytical reagent), hydrochloric acid (36–38%) and ammonia (analytical reagent) were purchased from Sinopharm Chemical Reagent Co. Ltd.

Preparation of the samples by the microwave-assisted hydrothermal method

In a typical experiment, an aqueous solution was prepared as follows: 0.1833, 0.3666, 0.7332 or 1.0998 g of BGP was dissolved in 20 mL deionized water, followed by the addition of 10 mL aqueous solution containing 0.1110, 0.2220, 0.4440 or 0.6660 g of anhydrous $CaCl_2$. The resulting solution was loaded into a modified polytetrafluoroethylene (TFM) cylindrical autoclave (volume 60 mL) with an outer diameter of 3.8 cm, inner diameter of 3.0 cm, and inner height of 9.3 cm. The autoclave was sealed, placed in a high-strength outer vessel, heated in a microwave oven to the temperature of 120, 140, 160 or 180 °C and was maintained at the desired temperature for 30 min. The microwave oven (frequency 2.45 GHz, maximum power 1000 W) was a microwave-hydrothermal synthesis system (MDS-6, Sineo, China) with a continuous heating mode. Temperature was measured with a platinum resistor temperature sensor (temperature range 0–250 °C) and controlled by automatically adjusting the power of microwave. The autoclave was rotated continuously during microwave irradiation for relatively uniform heating. After the termination of microwave heating and cooling to room temperature, the products were separated by centrifugation (10 000 rpm, 3 min), washed with deionized water three times, and freeze-dried. A control sample was prepared using a 30 mL aqueous solution containing 0.1833 g of BGP and 0.1110 g of anhydrous $CaCl_2$ with magnetic stirring for 20 min, and then the product was collected *via* centrifugation and washed with ethanol four times and dried in air at 60 °C.

Adsorption kinetics of metal ions

The amorphous calcium phosphate nanowires (ACPNWs), as the adsorbent for the removal of heavy metal ions, were prepared using a 30 mL aqueous solution containing 0.3666 g of BGP and 0.2220 g of anhydrous $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min. The adsorption kinetics for Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions onto the as-prepared ACPNWs were obtained from a series of experiments by dispersing 10 mg of ACPNWs into a 20 mL aqueous solution of Cd^{2+}, Cu^{2+}, Pb^{2+} or Zn^{2+} ions. The initial concentration of Cd^{2+}, Cu^{2+}, Pb^{2+} or Zn^{2+} ions was 100 mg L^{-1} in each aqueous solution with a pH value adjusted to 5.5 using diluted ammonia aqueous solution or dilute hydrochloric acid. The suspensions were magnetically stirred at room temperature for 5, 15, 30, 60, 120 or 180 min, and then 0.5 mL solution was withdrawn and diluted ten times with deionized water for inductively coupled plasma-optical emission spectrometry (ICP-OES) analysis. Data was represented as the mean value of three parallel experiments. The adsorption amount of heavy metal ions on the ACPNWs as the adsorbent was calculated by the following equation:

$$q = \frac{(C_0 - C_t) \times V}{m}$$

where q is the adsorption amount of metal ions on ACPNWs and C_0 (mg L^{-1}) and C_t (mg L^{-1}) are the concentrations of heavy metal ions before adsorption and after adsorption, respectively, at t min. V (L) is the volume of the solution containing heavy

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metal ions and m (mg) is the mass of ACPNWs. The respective adsorption isotherms of ACPNWs for Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions with five different concentrations of metal ions at 10, 50, 100, 200 and 300 mg L^{-1} in aqueous solution with a initial pH value of 5.5 were also studied.

Adsorption isotherms

The as-prepared ACPNWs, as the adsorbent for adsorption isotherm studies, were prepared using a 30 mL aqueous solution containing 0.3666 g of BGP and 0.2220 g of anhydrous $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min. The adsorption isotherms for Cd^{2+}, Cu^{2+}, Pb^{2+} or Zn^{2+} ions by ACPNWs were obtained by dispersing 5.0 mg of ACPNWs into a 20 mL aqueous solution containing heavy metal ions at room temperature. The initial concentration of heavy metal ions was 10, 50, 100, 200 or 300 mg L^{-1}. The initial pH value of each metal ion aqueous solution was adjusted to 5.5 using diluted ammonia aqueous solution or diluted hydrochloric acid. After 180 min of stirring, 0.5 mL solution was extracted and diluted for ICP-OES analysis. Data was represented as the mean value of three parallel experiments.

Selective metal ion adsorption of ACPNWs in aqueous solution

The ACPNWs, as the adsorbent for selective metal ion adsorption, were prepared using a 30 mL aqueous solution containing 0.3666 g of BGP and 0.2220 g of anhydrous $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min. The selective adsorption of the as-prepared ACPNWs was investigated in an aqueous solution containing a mixture of heavy metal ions. 10 mg of ACPNWs were dispersed in a 20 mL aqueous solution containing four types of heavy metal ions (Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+}) with a concentration of 10 mg L^{-1} for each of the metal ions. The initial pH value of metal ion aqueous solution was adjusted to 4.0, 5.5, 7.0 and 8.5 using dilute ammonia aqueous solution or hydrochloric acid. The suspensions were magnetically stirred at room temperature for 5, 15, 30, 60, 120 or 180 min, and then 0.5 mL supernatant was withdrawn and diluted for ICP-OES analysis. Data was represented as the mean value of three parallel experiments.

Characterization

Transmission electron microscopy (TEM) micrographs were obtained with a transmission electron microscope (TEM, Hitachi H-800, Japan); scanning electron microscopy (SEM) micrographs were obtained with a field-emission scanning electron microscope (SEM, Hitachi S-4800, Japan); thermogravimetric analysis (TG) was performed with a STA 409/PC simultaneous thermal analyzer (Netzsch, Germany) at a heating rate of 10 °C min^{-1} in flowing air; Brunauer–Emmett–Teller (BET) specific surface area and pore size distribution were measured with a surface area and pore size analyzer (V-Sorb 2800P, Gold APP, China); X-ray powder diffraction (XRD) patterns were recorded using an X-ray diffractometer (Rigaku D/max 2500V, Cu K_α radiation, $\lambda = 1.54178$ Å). The concentrations of metal ions were measured with inductively coupled plasma-optical emission spectrometry (ICP-OES, JY 2000-2, Horiba Scientific). Fourier transform infrared (FTIR) spectra were obtained on a FTIR spectrometer (FTIR-7600, Lamdba Scientific, Australia).

Results and discussion

The effect of the microwave hydrothermal temperature on the morphology and crystalline phase of the product was investigated. As shown in Fig. 1, the morphologies of the products prepared using an aqueous solution (30 mL) containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120, 140, 160 and 180 °C for 30 min were characterized by TEM. One can see that when the microwave hydrothermal temperature was 120 °C, the product consisted of nanowires (Fig. 1b); when the microwave hydrothermal temperature was increased to 140 °C, the product was composed of nanoparticles (Fig. 1c); when the microwave hydrothermal temperature was increased to 160 °C, the product consisted of nanoparticles and nanorods (Fig. 1d); when the microwave hydrothermal temperature was increased to 180 °C, the product consisted of nanorods (Fig. 1e).

The crystalline phases of the products prepared using an aqueous solution containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the microwave-assisted hydrothermal method at various temperatures for 30 min were characterized by XRD, which is shown in Fig. 2. The product prepared at 120 °C consisted of amorphous calcium phosphate (Fig. 2c). However, the products prepared at 140, 160 and 180 °C were composed of hydroxyapatite (Fig. 2d–f).

We performed a comparative investigation by preparing control samples using 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the conventional hydrothermal method instead of the microwave-assisted hydrothermal method at 120 °C for 30 min and 4 h. Fig. 1a shows the TEM micrograph of the control sample prepared using an aqueous solution (30 mL) containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the conventional hydrothermal method at 120 °C for 4 h instead of the microwave-assisted hydrothermal method. Our experiments showed that no product was obtained when the conventional hydrothermal time was 30 min at 120 °C. When the conventional hydrothermal time was 4 h, the product contained amorphous calcium phosphate (Fig. 2b), and nanowires were obtained (Fig. 1a). As discussed above, the ACPNWs are obtained by the microwave hydrothermal treatment for only 30 min (Fig. 1b). Therefore, compared with the conventional hydrothermal method, the microwave-assisted hydrothermal method exhibits its advantages such as rapidness, high efficiency and being energy saving.

The ACPNWs prepared using various amounts of BGP and $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min were characterized by TEM, as shown in Fig. 1b and f–h. As discussed above, the product consisted of nanowires when the amounts of BGP and $CaCl_2$ were 0.1833 g and 0.1110 g, respectively (Fig. 1b). When the amounts of BGP and $CaCl_2$ were increased to 0.3666 g and 0.2220 g, respectively, the product was still showed presence of nanowires (Fig. 1f). However, when the amounts of reactants were further increased, the sample

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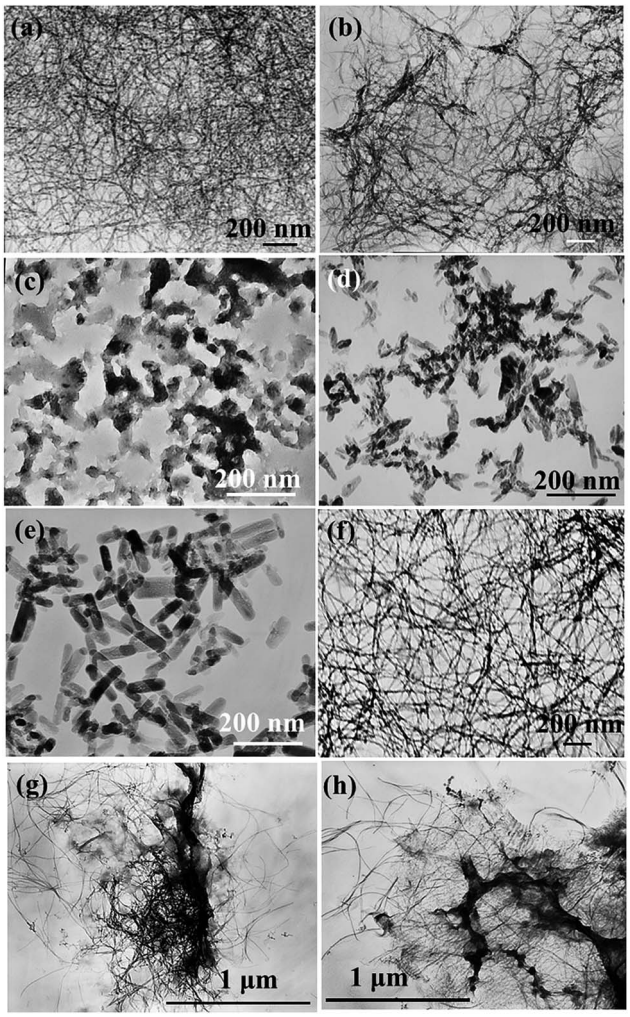

Fig. 1 TEM micrographs: (a) the control sample prepared using an aqueous solution (30 mL) containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the conventional hydrothermal method at 120 °C for 4 h; (b–e) the products prepared using an aqueous solution (30 mL) containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the microwave-assisted hydrothermal method at various temperatures for 30 min: (b) 120 °C, (c) 140 °C, (d) 160 °C and (e) 180 °C; (f–h) the products prepared using an aqueous solution (30 mL) containing various amounts of BGP and $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min: (f) 0.3666 g of BGP and 0.2220 g of $CaCl_2$, (g) 0.7332 g of BGP and 0.4440 g of $CaCl_2$, (h) 1.0998 g of BGP and 0.6660 g of $CaCl_2$.

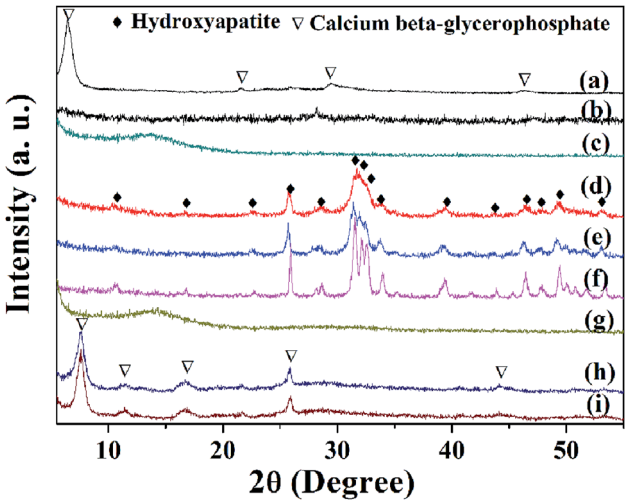

Fig. 2 XRD patterns: (a) the control sample prepared using an aqueous solution (30 mL) containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ with magnetic stirring for 20 min; (b) the control sample prepared using an aqueous solution (30 mL) containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the conventional hydrothermal method at 120 °C for 4 h; (c–f) the products prepared using an aqueous solution (30 mL) containing 0.1833 g of BGP and 0.1110 g of $CaCl_2$ by the microwave-assisted hydrothermal method at various temperatures for 30 min: (c) 120 °C, (d) 140 °C, (e) 160 °C and (f) 180 °C; (g–i) the products prepared using an aqueous solution (30 mL) containing various amounts of BGP and $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min: (g) 0.3666 g of BGP and 0.2220 g of $CaCl_2$, (h) 0.7332 g of BGP and 0.4440 g of $CaCl_2$, (i) 1.0998 g of BGP and 0.6660 g of $CaCl_2$.

exhibited nanowires as the main product and nanoparticles as the minor product.

Fig. 2a, b and g–i show the XRD patterns of the control samples and the products prepared using various amounts of BGP and $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min. When the amounts of BGP and $CaCl_2$ were 0.1833 g and 0.1110 g, or 0.3666 g and 0.2220 g, respectively, the amorphous calcium phosphate nanowires were formed at 120 °C for 30 min (Fig. 2c and g). When the amounts of BGP and $CaCl_2$ were further increased, the product contained a crystalline phase of β-glycerophosphate calcium salt (Fig. 2h and i).[38] The control sample obtained without microwave hydrothermal treatment also consisted of a crystalline β-glycerophosphate calcium salt (Fig. 2a). The largest XRD peak of two β-glycerophosphate calcium salt samples is located at approximately 2θ angles of 6.6° and 7.4°, respectively; this difference may be explained by the different degree of hydration of β-glycerophosphate calcium salt.

We propose a possible formation mechanism for the ACPNWs (Fig. 3). When calcium ions are added into a BGP aqueous solution, calcium ions react with BGP molecules to produce β-glycerophosphate calcium salt. β-Glycerophosphate calcium salt acts as the precursor for the synthesis of ACPNWs. Some BGP molecules can hydrolyze to form phosphate ions under the microwave hydrothermal conditions, and then phosphate ions react with calcium ions to form calcium phosphate nuclei, which then grow to form amorphous calcium phosphate nanowires, and the remaining unhydrolyzed BPG molecules can act as a stabilizer to help in the stabilization of the amorphous phase of calcium phosphate at 120 °C to inhibit the transformation of amorphous calcium phosphate to hydroxyapatite in aqueous solution. This result is different from that observed for ACP nanospheres prepared using ATP as an organic phosphate source in our previous study.[23] At higher microwave hydrothermal temperatures (140, 160 or 180 °C), BGP molecules may completely hydrolyze and amorphous calcium phosphate may transform to hydroxyapatite (Fig. 2d–f).

Fig. 4a–c display the SEM micrographs, TG curve and nitrogen adsorption–desorption isotherm of the ACPNWs prepared using 0.3666 g of BGP and 0.2220 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min.

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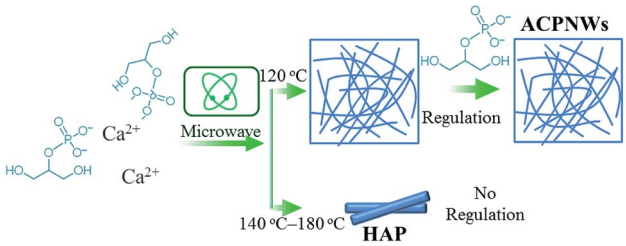

Fig. 3 Schematic illustration of the processes applied for the preparation of products using BGP and $CaCl_2$ by the microwave-assisted hydrothermal method.

Fig. 4a shows that the product consists of amorphous calcium phosphate nanowires, which is consistent with the TEM observation. The SEM image in the inset of Fig. 1a further exhibits the morphology of the nanowires with diameters of tens of nanometers. The TG curve of the ACPNWs shows the residual weight to be 55.5% at 1200 °C, the decreased weight may be explained by the existence of β-glycerophosphate and hydrated water in the product. According to the International Union of Pure and Applied Chemistry (IUPAC), the nitrogen adsorption–desorption isotherm of the as-synthesized ACPNWs can be classified as a type IV isotherm loop, which has the characteristics of porous structures.[39] The BET specific surface area of ACPNWs is 115 $m^2\ g^{-1}$, and the average Barrett–Joyner–Halenda (BJH) pore size of ACPNWs is 24 nm.

The potential application of the as-prepared ACPNWs as the adsorbent in water treatment using typical heavy metal ions, including Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+}, was investigated. The ACPNWs were prepared using 0.3666 g of BGP and 0.2220 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min. The respective adsorption kinetics of the ACPNWs for Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions are displayed in Fig. 5a–d. The initial concentration of Cd^{2+}, Cu^{2+}, Pb^{2+} or Zn^{2+} was 100 mg L^{-1} and the initial pH value was adjusted to 5.5. The adsorption for four heavy metals nearly reached a plateau after 5 min, implying that the adsorption of ACPNWs for heavy metals was a fast process. ICP analysis results showed that the removal percentages of Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} were 53.6%, 78.4%, 98.4% and 39.7%, respectively, after 180 min. The adsorption capabilities of ACPNWs for 100 mg L^{-1} Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} were 107, 157, 197 and 79.4 mg g^{-1}, respectively, in an aqueous solution with an initial pH value of 5.5. In consideration of the various molar weights, the adsorption molar amounts of Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} onto ACPNWs were 0.955, 2.47, 0.951 and 1.21 mmol g^{-1}, respectively. The different adsorption efficiency of ACPNWs for metal ions can be explained by the difference of ionic radii of the metals ions during adsorption.[40] The ionic radii of Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions with a coordination number of 6 are 95, 73, 119 and 74 pm,[41] indicating that metal ions with a small ionic radius are preferably adsorbed onto the ACPNWs. In addition, compared with the hydroxyapatite sheet-assembled microspheres with microsized pores,[7] the ACPNWs had a faster and higher adsorption for heavy metal ions, especially Pb^{2+} ions. This may be attributed to the nanosized pores and a relatively high specific surface area (115 $m^2\ g^{-1}$) of the ACPNWs, which facilitated the adsorption process.

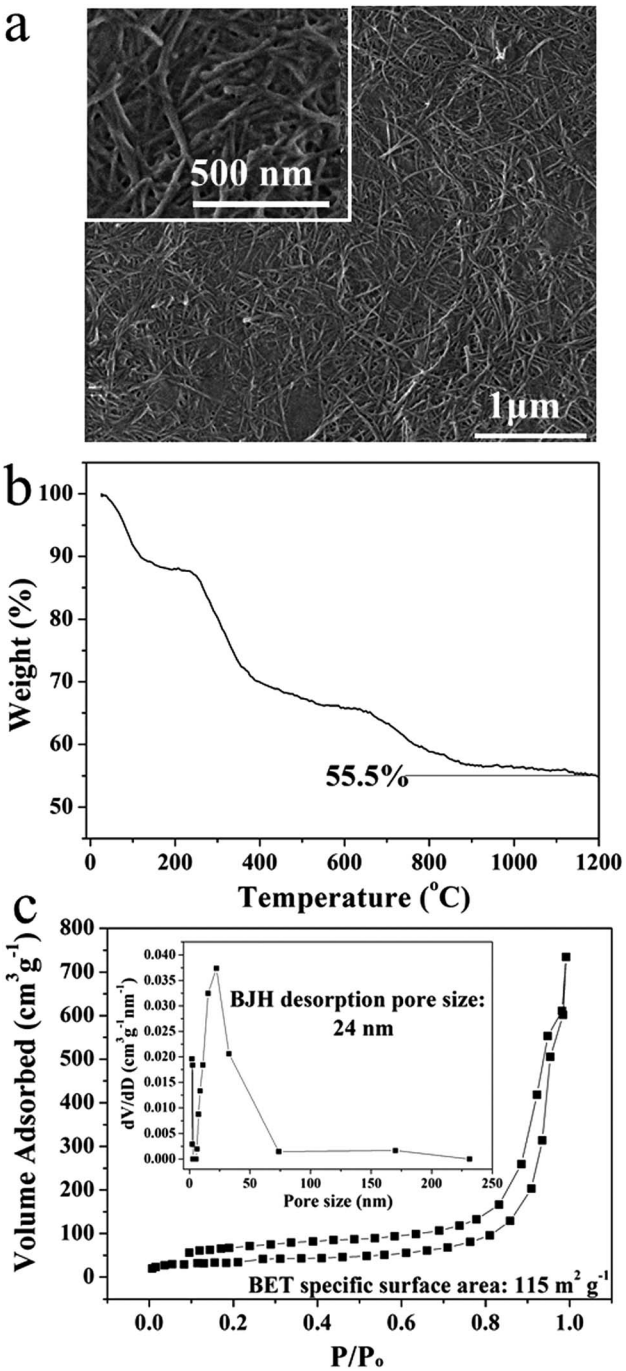

Fig. 4 (a) SEM micrograph, (b) TG curve and (c) nitrogen adsorption–desorption isotherm of the as-synthesized ACPNWs prepared using an aqueous solution containing 0.3666 g of BGP and 0.2220 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min. The inset in (a) shows a SEM micrograph with a larger magnification, and the inset in (b) shows the pore size distribution of the ACPNWs.

The following pseudo second-order equation was used to describe the kinetics of metal adsorption process on the ACPNWs adsorbent in aqueous solution:

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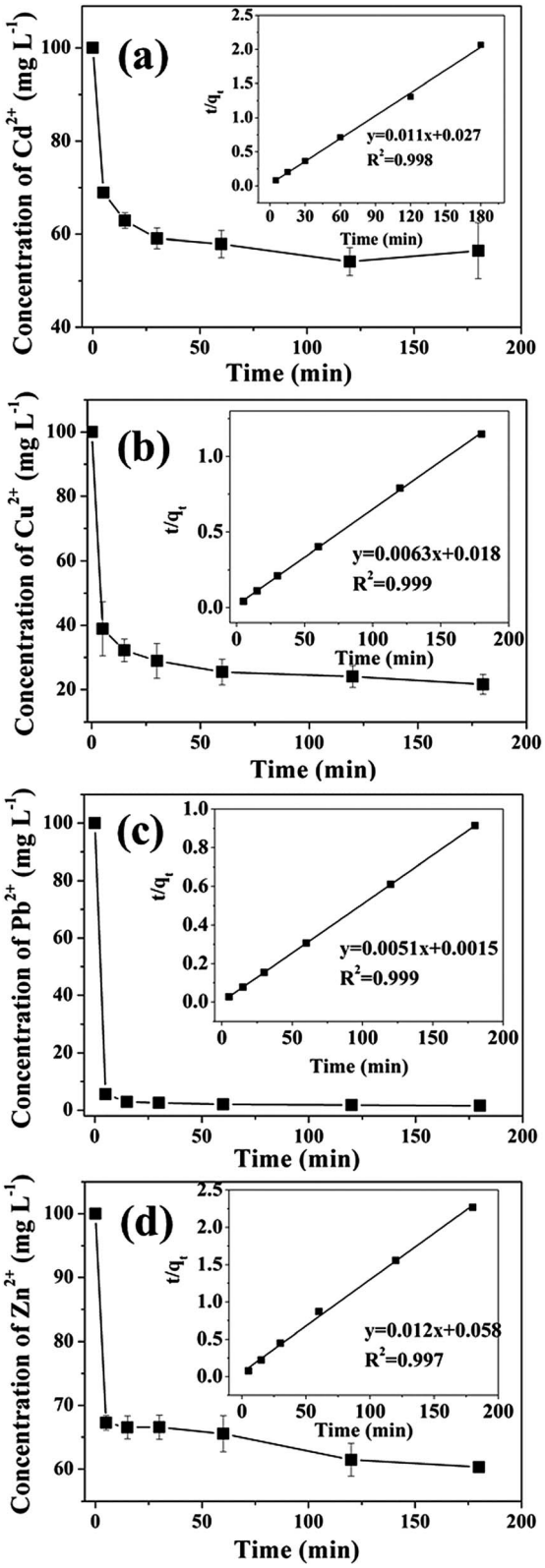

Fig. 5 Adsorption kinetics of Cd^{2+} (a), Cu^{2+} (b), Pb^{2+} (c) and Zn^{2+} (d) ions on the ACPNWs at an initial metal ion concentration of 100 mg L^{-1} and an initial pH value of 5.5. The ACPNWs were prepared using 0.3666 g of BGP and 0.2220 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min. The insets in (a)–(d) show the relationship between t/q_t and adsorption time (t).

$$\frac{t}{q_t} = \frac{1}{kq_e^2} + \frac{t}{q_e}$$

where k (g mg^{-1} min^{-1}) is the second-order rate constant, q_t (mg g^{-1}) is the adsorption amount of metal ions at time t (min), q_e (mg g^{-1}) is the equilibrium adsorption amount of metal ions, and t (min) is the adsorption time.

Fig. 5 shows the concentrations of metal ions (Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+}) in solution *versus* the adsorption time. One can see that the concentration of metal ions in solution decreased with the adsorption time, and the concentration of metal ions decreased rapidly at an early stage, implying the rapid adsorption of metal ions onto the ACPNWs adsorbent. It should be noted that the adsorption of Pb^{2+} ions was especially efficient; Pb^{2+} ions were almost completely adsorbed in a short period of time (about 5 min). In the insets of Fig. 5a–d, a straight line can be fitted with a slope of $1/q_e$ by plotting t/q_t *versus* t. The correlation coefficients R^2 are 0.998, 0.999, 0.999 and 0.997, respectively, implying that the adsorption kinetics of Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions on the ACPNWs could be well described by the pseudo second-order equation.

Fig. 6 displays the XRD patterns of the as-prepared ACPNWs after adsorption for heavy metal ions (Cd^{2+}, Cu^{2+}, Pb^{2+} or Zn^{2+}) in an aqueous solution with an initial concentration of 100 mg L^{-1} and an initial pH value of 5.5. One can see that after the adsorption of Cd^{2+}, Cu^{2+}, Pb^{2+} or Zn^{2+} ions for 180 min, the final products were still amorphous. However, a broad and weak peak appeared at around 30° in the XRD pattern of the product after the adsorption of Pb^{2+} ions on the ACPNWs for 180 min, indicating the formation of poorly crystallized lead phosphate.

Furthermore, Fig. 7 shows the FTIR spectra of BGP, ACPNWs and the products after the adsorption of Cd^{2+}, Cu^{2+}, Pb^{2+} or Zn^{2+} ions on the ACPNWs for 180 min. According to another study,[42] the broad peak between 3600 cm^{-1} and 3100 cm^{-1} and the peak at about 1630 cm^{-1} are assigned to the hydration of the products; the small absorption bands at about 2964 and 2915 cm^{-1} may be attributed to the stretching vibration of $-CH_2$ groups from BGP molecules; the adsorption peaks at around 1083, 570

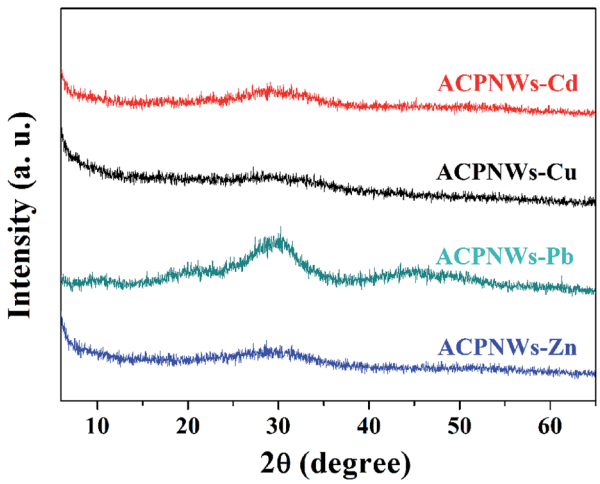

Fig. 6 XRD patterns of the products after the respective adsorption of ACPNWs for Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions for 180 min.

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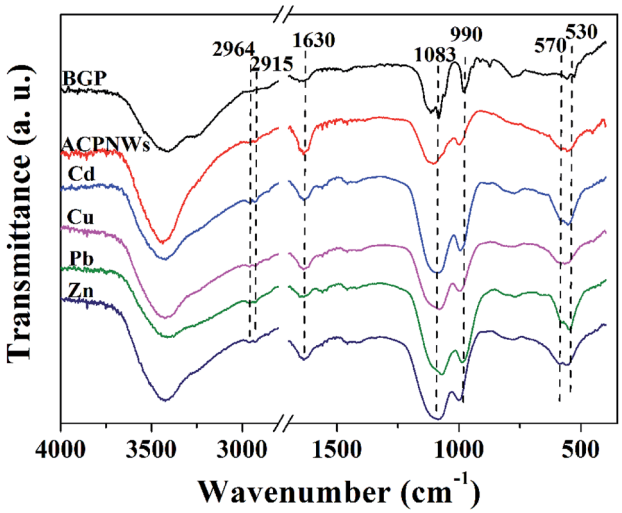

Fig. 7 FTIR spectra of (a) BGP, (b) as-prepared ACPNWs and the products after the respective adsorption of ACPNWs for Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions for 180 min. ACPNWs were prepared using 0.3666 g of BGP and 0.2220 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min.

and 530 cm^{-1} are characteristic bands of the PO_4^{3-} group; and the absorption peak around 990 cm^{-1} is the characteristic band of P–O–C group from BGP molecules. The results indicate the existence of BGP molecules in the as-prepared ACPNWs and the products after adsorption of heavy metal ions. The existence of BGP molecules can help stabilize the amorphous calcium phosphate in the products.

During the adsorption of heavy metal ions, the immobilization of metal ions on the adsorbent occurs through a three-step mechanism: rapid surface complexation, partial dissolution of calcium phosphate and metal ion exchange with Ca^{2+} ions, resulting in the formation of a calcium/metal phosphate salt.[43–45] Compared with the crystalline calcium phosphate, the

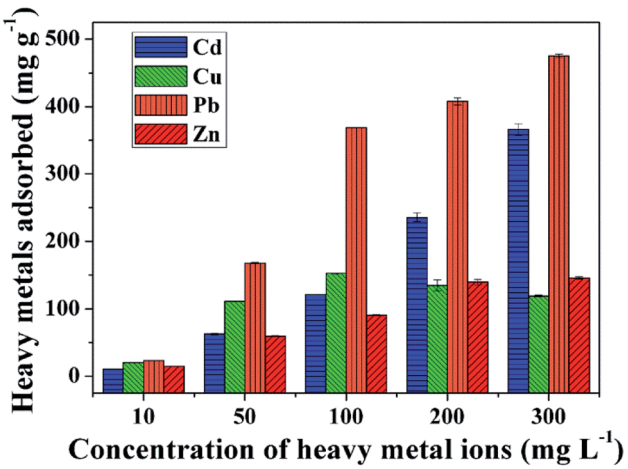

Fig. 8 Adsorption performance of ACPNWs for Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions at different initial concentrations and an initial pH value of 5.5. The ACPNWs were prepared using 0.3666 g of BGP and 0.2220 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min.

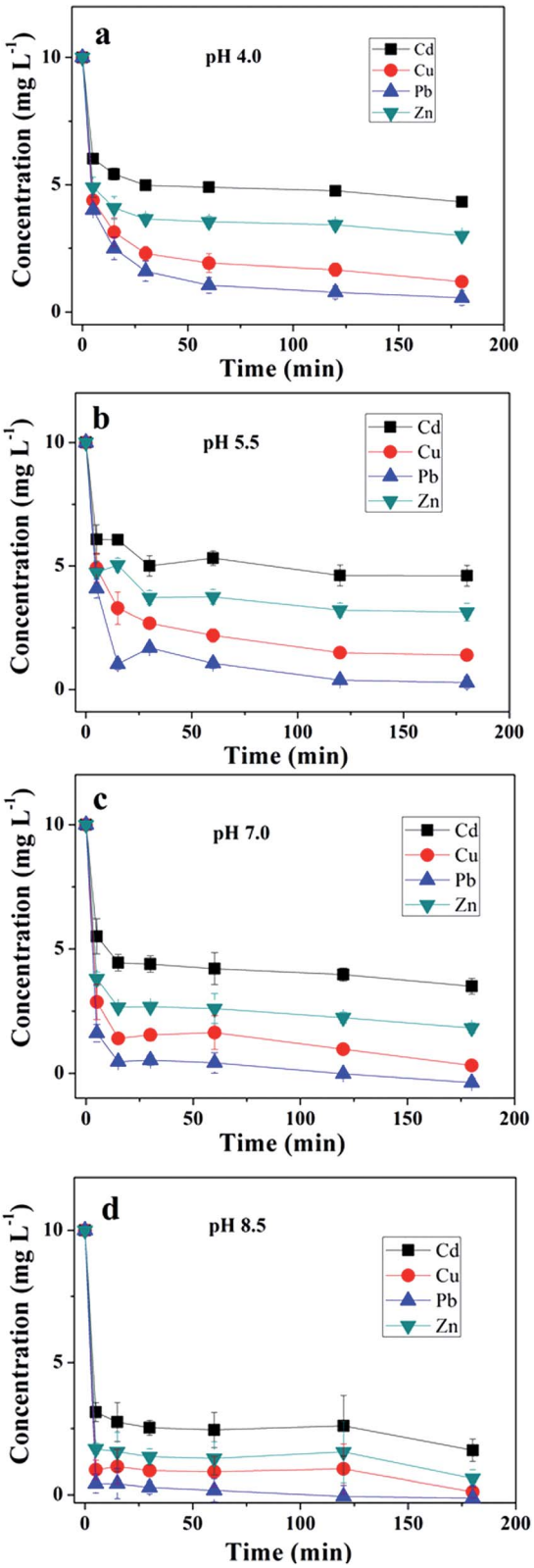

Fig. 9 Selective adsorption behavior of the ACPNWs for heavy metal ions in aqueous solution at various pH values: (a) 4.0; (b) 5.5; (c) 7.0; (d) 8.5. The ACPNWs were prepared using 0.3666 g of BGP and 0.2220 g of $CaCl_2$ by the microwave-assisted hydrothermal method at 120 °C for 30 min.

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amorphous calcium phosphate nanowires can easily exchange calcium ions with heavy metal ions; this can explain the fast metal ion adsorption behavior of the ACPNWs as the adsorbent. Furthermore, the BGP molecules adsorbed on ACPNWs may play an important role in the adsorption of metal ions. During the rapid surface complexation, the BGP molecules adsorbed onto the ACPNWs may help to attract heavy metal ions due to the electrostatic force between phosphate groups and metal ions. After the ion exchange between Ca^{2+} and heavy metal ions, the BGP molecules may also prevent the crystallization of calcium/metal phosphate salt.

The respective adsorption performance of ACPNWs for Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions with five different concentrations of metal ions at 10, 50, 100, 200 and 300 mg L^{-1} in an aqueous solution with a pH value of 5.5 were also studied and are shown in Fig. 8. The experiments indicate that the adsorption amount of metal ions on the ACPNWs increased with increase in the initial concentration of metal ions, except for Cu^{2+} ions with concentrations of 200 mg L^{-1} and higher. When the concentration of Cu^{2+} ions was over 200 mg L^{-1}, the precipitation of $Cu(OH)_2$ occurred. The initial high concentration of metal ions provides a driving force for the metal ions to transfer from the aqueous phase to the adsorbent. Therefore, a high initial concentration of metal ions can enhance metal ion adsorption on the ACPNWs. The adsorption amounts of Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} ions on the ACPNWs were 366, 119, 475 and 145 mg g^{-1}, respectively, at an initial concentration of 300 mg L^{-1}. Thus, the as-prepared ACPNWs have high adsorption ability for heavy metal ions and are promising for applications in water treatment.

The selective adsorption behavior of the ACPNWs for heavy metal ions was investigated using an aqueous solution containing four metal ions of Cd^{2+}, Cu^{2+}, Pb^{2+} and Zn^{2+} with a concentration of 10 mg L^{-1} for each metal ion at different pH values of 4.0, 5.5, 7.0 and 8.5. Our experiments showed that the pH value of the solution strongly affected the adsorption of heavy metal ions on the ACPNWs. Fig. 9 shows the selective adsorption behavior of the ACPNWs for heavy metal ions at various pH values. It displays a trend that the adsorption capability and adsorption rate of four metal ions on the ACPNWs increased with increasing pH value. At a higher pH value, the solubility of the ACPNWs decreased, promoting the surface adsorption and complexation of heavy metal ions for better immobilization. The adsorption rates of the four metal ions on the ACPNWs were high in the aqueous solutions with various initial pH values, especially at the early stage. The ACPNWs also showed a high selective adsorption behavior for Pb^{2+} ions. Pb^{2+} ions were almost completely adsorbed in a short period of time at different pH values.

Conclusions

Amorphous calcium phosphate nanowires (ACPNWs) have been successfully prepared using $CaCl_2$ as a calcium source and β-glycerophosphate disodium salt (BGP) as a phosphorus source by the microwave-assisted hydrothermal method. The microwave hydrothermal temperature and amounts of BGP and $CaCl_2$ were important parameters for controlling the morphology and crystalline phase of the products. This preparation strategy is facile, rapid, energy-saving, and environment-friendly. The possible formation mechanism of the ACPNWs was proposed. The as-prepared ACPNWs have a large specific surface area and were explored for potential application in the removal of heavy metal ions. The experimental results indicated that the as-prepared ACPNWs had high adsorption capability for heavy metal ions and high selective adsorption ability for Pb^{2+} ions, and therefore they are a promising adsorbent for the removal of heavy metal ions from water.

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